LAPORAN LATIHAN PRAKTIKUM JOBSHEET 5
BRUTE FORCE & DIVIDE CONQUER

Nama : Jonathan Abdiel Haryono
Absen : 14
Kelas : TI 1G

LATIHAN PRAKTIKUM

1. Sebuah kampus memiliki daftar nilai mahasiswa dengan data sesuai tabel di bawah ini

Nama	NIM	Tahun Masuk	Nilai UTS	Nilai UAS
Ahmad	220101001	2022	78	82
Budi	220101002	2022	85	88
Cindy	220101003	2021	90	87
Dian	220101004	2021	76	79
Eko	220101005	2023	92	95
Fajar	220101006	2020	88	85
Gina	220101007	2023	80	83
Hadi	220101008	2020	82	84

Tentukan:

a) Nilau UTS tertinggi tertinggi menggunakan Divide and Conquer!

b) Nilai UTS terendah menggunakan Divide and Conquer!

c) Rata-rata nilai UAS dari semua mahasiswa menggunakan Brute Force!

JAWABAN

```
package Minggu5.BruteForceDivideConquer;


public class LatihanPraktikum {

    static String[][] mhs = {

        {"Ahmad", "220101001", "2022"},

        {"Budi", "220101002", "2022"},

        {"Cindy", "220101003", "2021"},

        {"Dian", "220101004", "2021"},

        {"Eko", "220101005", "2023"},

        {"Fajar", "220101006", "2020"},
```

```
    {"Gina", "220101007", "2023"},
    {"Hadi", "220101008", "2020"}

};

static int[] nilaiUTS = {78, 85, 90, 76, 92, 88, 80, 82};

static int[] nilaiUAS = {82, 88, 87, 79, 95, 85, 83, 84};

static double highest(int nilai[], int l, int r) {

    int mid = (l + r) / 2;
    double highest, highestL, hightestR;



    if (r == l + 1) {
        if (nilai[l] >= nilai[r]) {
            highest = nilai[l];
        }
        else {
            highest = nilai[r];
        }
    }
    else if (r == l) {
        highest = nilai[l];
        return highest;
```

```
    }

    highestL = highest(nilai, l, mid);
    hightestR = highest(nilai, mid + 1, r);

    if (highestL > hightestR) {
        highest = highestL;
    }
    else {
        highest = hightestR;
    }

    return highest;
}

static double lowest(int nilai[], int l, int r) {

    int mid = (l + r) / 2;
    double lowest, lowestL, lowestR;


    if (r == l + 1) {
        if (nilai[l] <= nilai[r]) {
            lowest = nilai[l];
        }
        else {
```

```
            lowest = nilai[r];
        }
    }
    else if (r == l) {
        lowest = nilai[l];
        return lowest;
    }

    lowestL = lowest(nilai, l, mid);
    lowestR = lowest(nilai, mid + 1, r);

    if (lowestL < lowestR) {
        lowest = lowestL;
    }
    else {
        lowest = lowestR;
    }

    return lowest;
}

static double average(int nilai[]) {
    double res = 0;

    for (int i = 0; i < nilai.length; i++) {
        res += nilai[i];
```

```java
        }

        return res / nilai.length;

    }

    public static void main(String[] args) {
        System.out.println("Nilai UTS Tertinggi : " + highest(nilaiUTS, 0, nilaiUTS.length-1));
        System.out.println("Nilai UAS Terendah : " + lowest(nilaiUTS, 0, nilaiUTS.length-1));

        System.out.println();

        System.out.println("Nilai UTS Tertinggi : " + highest(nilaiUAS, 0, nilaiUTS.length-1));
        System.out.println("Nilai UAS Terendah : " + lowest(nilaiUAS, 0, nilaiUTS.length-1));

        System.out.println("\n----------------------------\n");

        System.out.println("Nilai rata-rata UAS seluruh mahasiswa : " + average(nilaiUAS) + "\n");
    }
}
```

```
Nilai UTS Tertinggi : 92.0
Nilai UAS Terendah : 76.0

Nilai UTS Tertinggi : 95.0
Nilai UAS Terendah : 79.0

----------------------------

Nilai rata-rata UAS seluruh mahasiswa : 85.375
```